

Лабораторная работа №04

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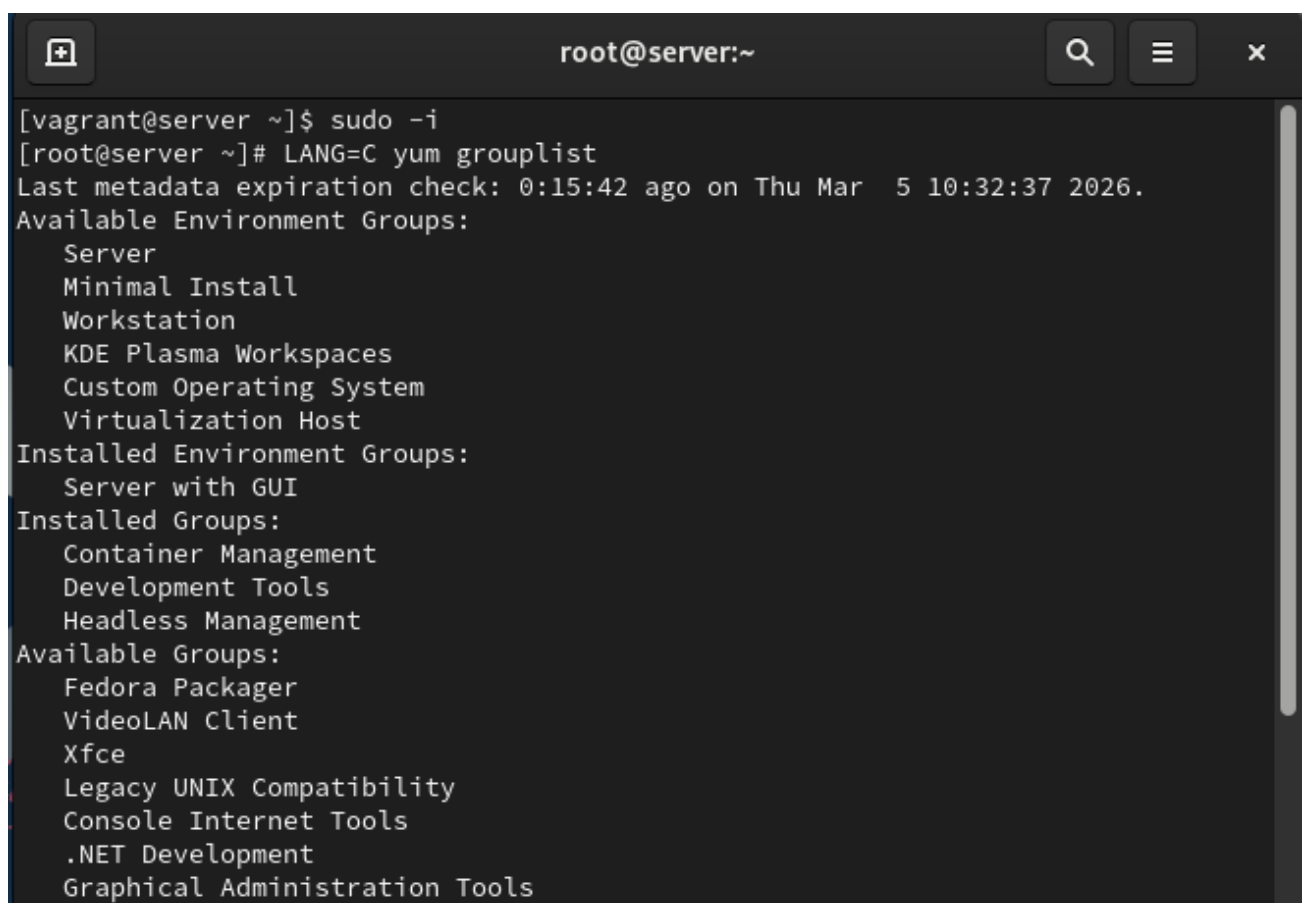
Лабораторная работа №04

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Цель работы

Приобретение практических навыков администрирования сетевых подсистем

Загрузим нашу операционную систему и перейдём в рабочий каталог с проектом. Далее запустим виртуальную машину server (рис. @fig-1):



```
root@server:~  
[vagrant@server ~]$ sudo -i  
[root@server ~]# LANG=C yum grouplist  
Last metadata expiration check: 0:15:42 ago on Thu Mar  5 10:32:37 2026.  
Available Environment Groups:  
  Server  
  Minimal Install  
  Workstation  
  KDE Plasma Workspaces  
  Custom Operating System  
  Virtualization Host  
Installed Environment Groups:  
  Server with GUI  
Installed Groups:  
  Container Management  
  Development Tools  
  Headless Management  
Available Groups:  
  Fedora Packager  
  VideoLAN Client  
  Xfce  
  Legacy UNIX Compatibility  
  Console Internet Tools  
  .NET Development  
  Graphical Administration Tools
```

Рис. 1: Этап 1

```
[root@server ~]# dnf -y groupinstall "Basic Web Server"
Last metadata expiration check: 0:16:28 ago on Thu 05 Mar 2026 10:32:37 AM UTC.
Dependencies resolved.
=====
Package                Arch      Version              Repository           Size
=====
Installing group/module packages:
httpd                  x86_64    2.4.62-7.el9_7.3     appstream            45 k
httpd-manual           noarch    2.4.62-7.el9_7.3     appstream            2.2 M
mod_fcgid              x86_64    2.3.9-28.el9         appstream            74 k
mod_ssl                x86_64    1:2.4.62-7.el9_7.3   appstream            110 k
Installing dependencies:
apr                    x86_64    1.7.0-12.el9_3       appstream            122 k
apr-util              x86_64    1.6.1-23.el9         appstream            94 k
apr-util-bdb          x86_64    1.6.1-23.el9         appstream            12 k
httpd-core            x86_64    2.4.62-7.el9_7.3     appstream            1.4 M
httpd-filesystem      noarch    2.4.62-7.el9_7.3     appstream            12 k
httpd-tools           x86_64    2.4.62-7.el9_7.3     appstream            79 k
rocky-logos-httpd     noarch    90.16-1.el9          appstream            24 k
=====
```

Рис. 2: Этап 2

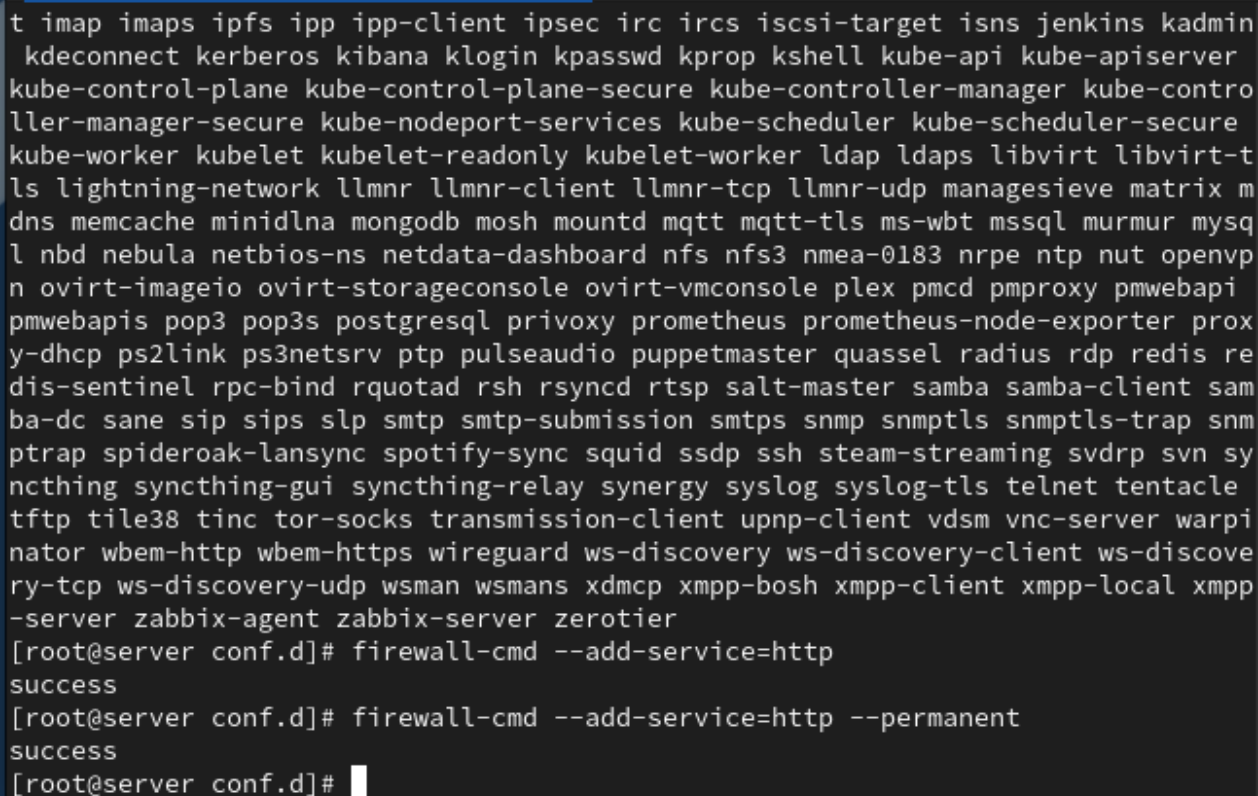
На виртуальной машине **server** войдём под нашим пользователем и откроем терминал. Перейдём в режим суперпользователя и установим **dhcp-server** (рис. @fig-2):

Скопируем файл примера конфигурации ДНСП **dhcpd.conf.example** из каталога **/usr/share/doc/dhcp*** в каталог **/etc/dhcp** и переименуем его в файл с названием **dhcpd.conf** (рис. @fig-3):

```
Complete!
[root@server ~]# cd /etc/httpd/conf и /etc/httpd/conf.d
-bash: cd: too many arguments
[root@server ~]# cd ..
[root@server /]# cd /etc/httpd/conf и /etc/httpd/conf.d
-bash: cd: too many arguments
[root@server /]# cd /etc/httpd/conf
[root@server conf]# ls
httpd.conf  magic
[root@server conf]# cd /etc/httpd/conf.d
[root@server conf.d]# ls
autoindex.conf  manual.conf  rocky-snipolicy.conf  userdir.conf
fcgid.conf      README      ssl.conf              welcome.conf
[root@server conf.d]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client imap imaps pop3 pop3s smtp ssh
[root@server conf.d]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp am
qps apcupsd audit ausweisapp2 bacula bacula-client bareos-director bareos-fileda
emon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-r
pc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit col
lectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dh
```

Рис. 3: Этап 3

Откроем файл /etc/dhcp/dhcpd.conf на редактирование. В этом файле заменим строки option domain-name и option domain-name-servers, раскомментируем строку authoritative и на базе одного из приведённых в файле примеров конфигурирования подсети зададим собственную конфигурацию DHCP-сети (рис. @fig-4):



```
t imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin
kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver
kube-control-plane kube-control-plane-secure kube-controller-manager kube-contro
ller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure
kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-t
ls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix m
dns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysq
l nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvp
n ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi
pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox
y-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis re
dis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client sam
ba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snm
ptrap spideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn sy
ncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle
tftp tile38 tinc tor-socks transmission-client upnp-client vdsm vnc-server warpi
nator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discove
ry-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp
-server zabbix-agent zabbix-server zerotier
[root@server conf.d]# firewall-cmd --add-service=http
success
[root@server conf.d]# firewall-cmd --add-service=http --permanent
success
[root@server conf.d]#
```

Рис. 4: Этап 4

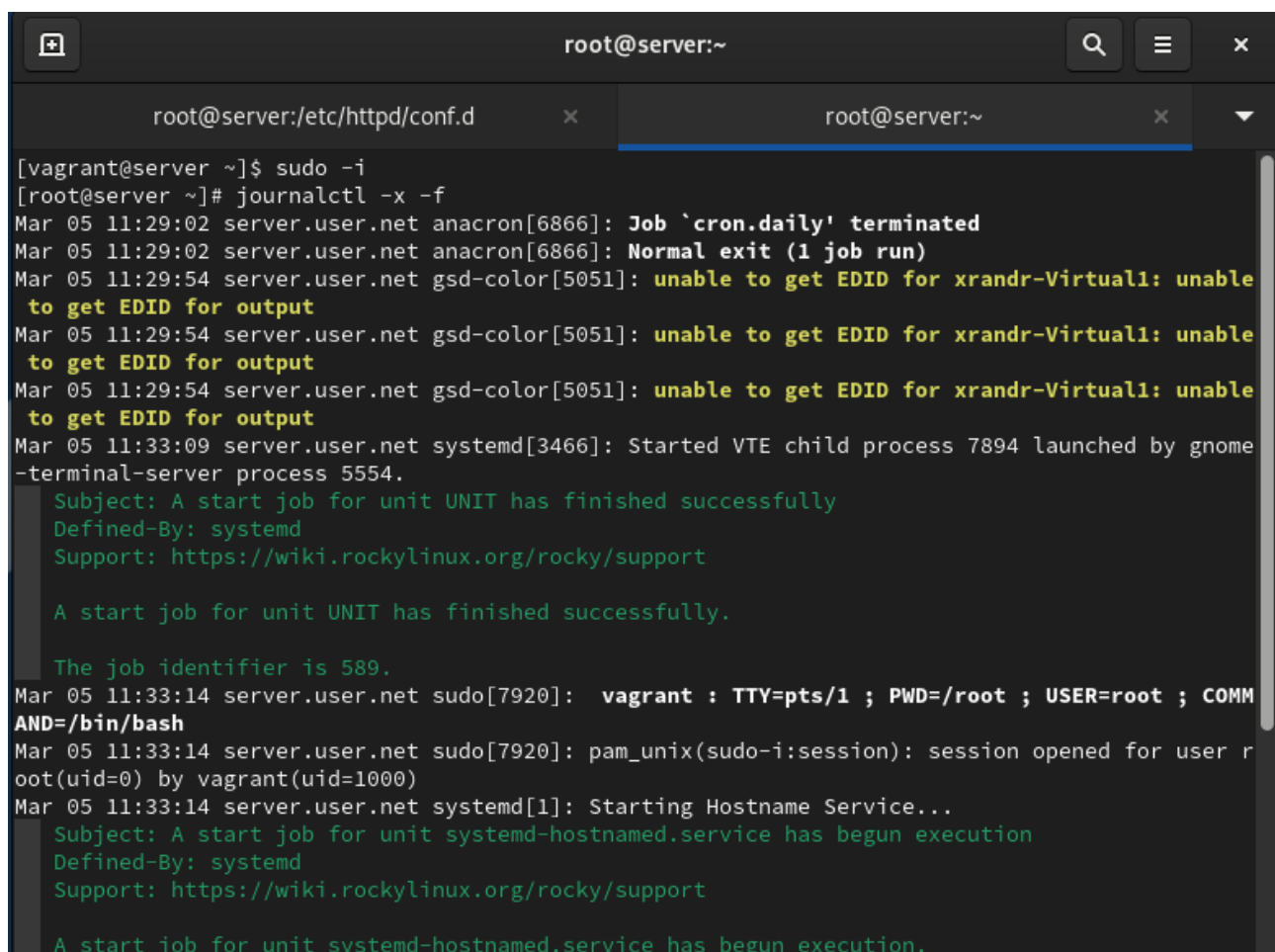
Откроем файл /etc/systemd/system/dhcpd.service на редактирование и заменим в нём строку, указав интерфейс eth1. Затем перезагрузим конфигурацию systemd и разрешим загрузку DHCP-сервера при запуске виртуальной машины server (рис. @fig-5):

Добавим запись для DHCP-сервера в конце файла прямой DNS-зоны /var/named/master/master.zone и в конце файла обратной зоны /var/named/master/rz/192.168.1. Перезапустим named и проверим, что можно обратиться к DHCP-серверу по имени (рис. @fig-6):

Внесём изменения в настройки межсетевого экрана узла server, разрешив работу с DHCP, и восстановим контекст безопасности в SELinux (рис. @fig-7):

Этап выполнения работы №8

Этап выполнения работы №9



A terminal window titled 'root@server:~' with a search icon, a menu icon, and a close icon in the top right. Below the title bar, there are two tabs: 'root@server:/etc/httpd/conf.d' and 'root@server:~'. The terminal content shows the following:

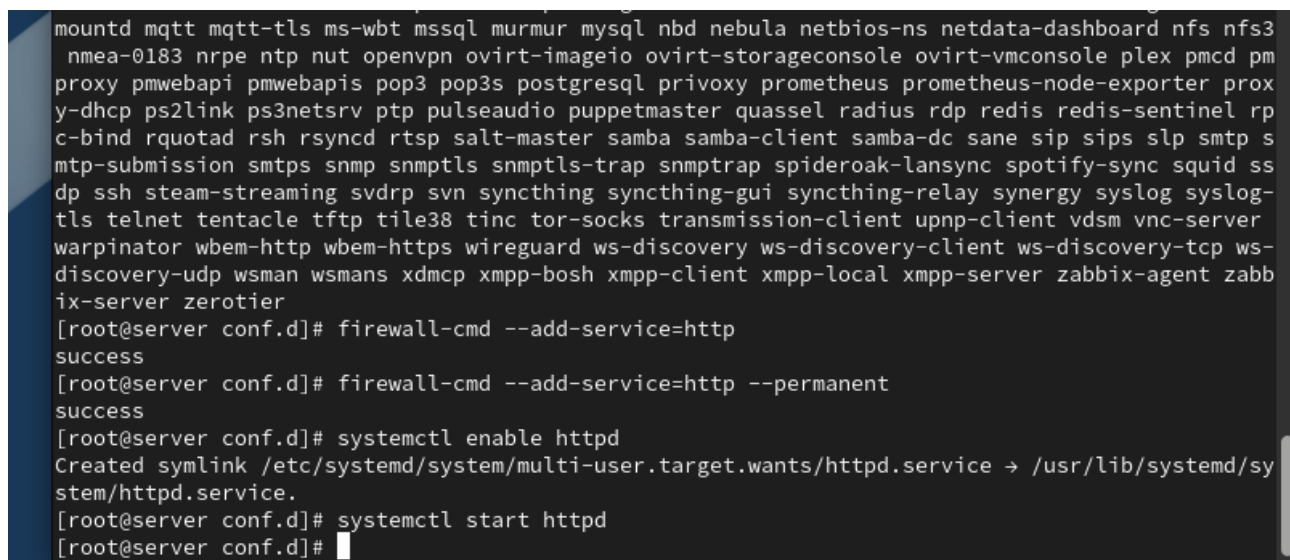
```
[vagrant@server ~]$ sudo -i
[root@server ~]# journalctl -x -f
Mar 05 11:29:02 server.user.net anacron[6866]: Job `cron.daily' terminated
Mar 05 11:29:02 server.user.net anacron[6866]: Normal exit (1 job run)
Mar 05 11:29:54 server.user.net gsd-color[5051]: unable to get EDID for xrandr-Virtual1: unable
to get EDID for output
Mar 05 11:29:54 server.user.net gsd-color[5051]: unable to get EDID for xrandr-Virtual1: unable
to get EDID for output
Mar 05 11:29:54 server.user.net gsd-color[5051]: unable to get EDID for xrandr-Virtual1: unable
to get EDID for output
Mar 05 11:33:09 server.user.net systemd[3466]: Started VTE child process 7894 launched by gnome
-terminal-server process 5554.
Subject: A start job for unit UNIT has finished successfully
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit UNIT has finished successfully.

The job identifier is 589.
Mar 05 11:33:14 server.user.net sudo[7920]: vagrant : TTY=pts/1 ; PWD=/root ; USER=root ; COMM
AND=/bin/bash
Mar 05 11:33:14 server.user.net sudo[7920]: pam_unix(sudo-i:session): session opened for user r
oot(uid=0) by vagrant(uid=1000)
Mar 05 11:33:14 server.user.net systemd[1]: Starting Hostname Service...
Subject: A start job for unit systemd-hostnamed.service has begun execution
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit systemd-hostnamed.service has begun execution.
```

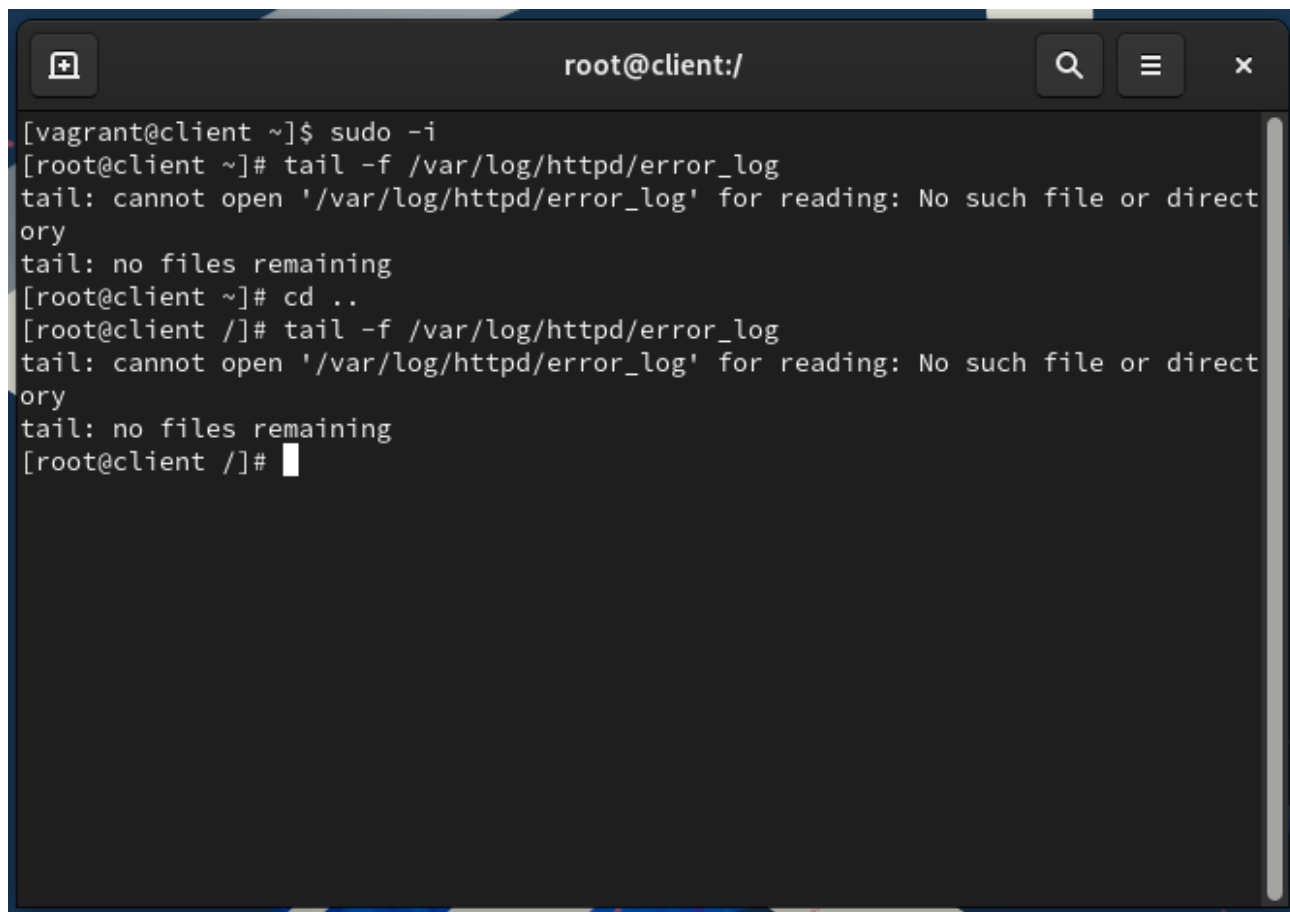
Рис. 5: Этап 5



A terminal window showing a list of services and then configuring httpd. The content is as follows:

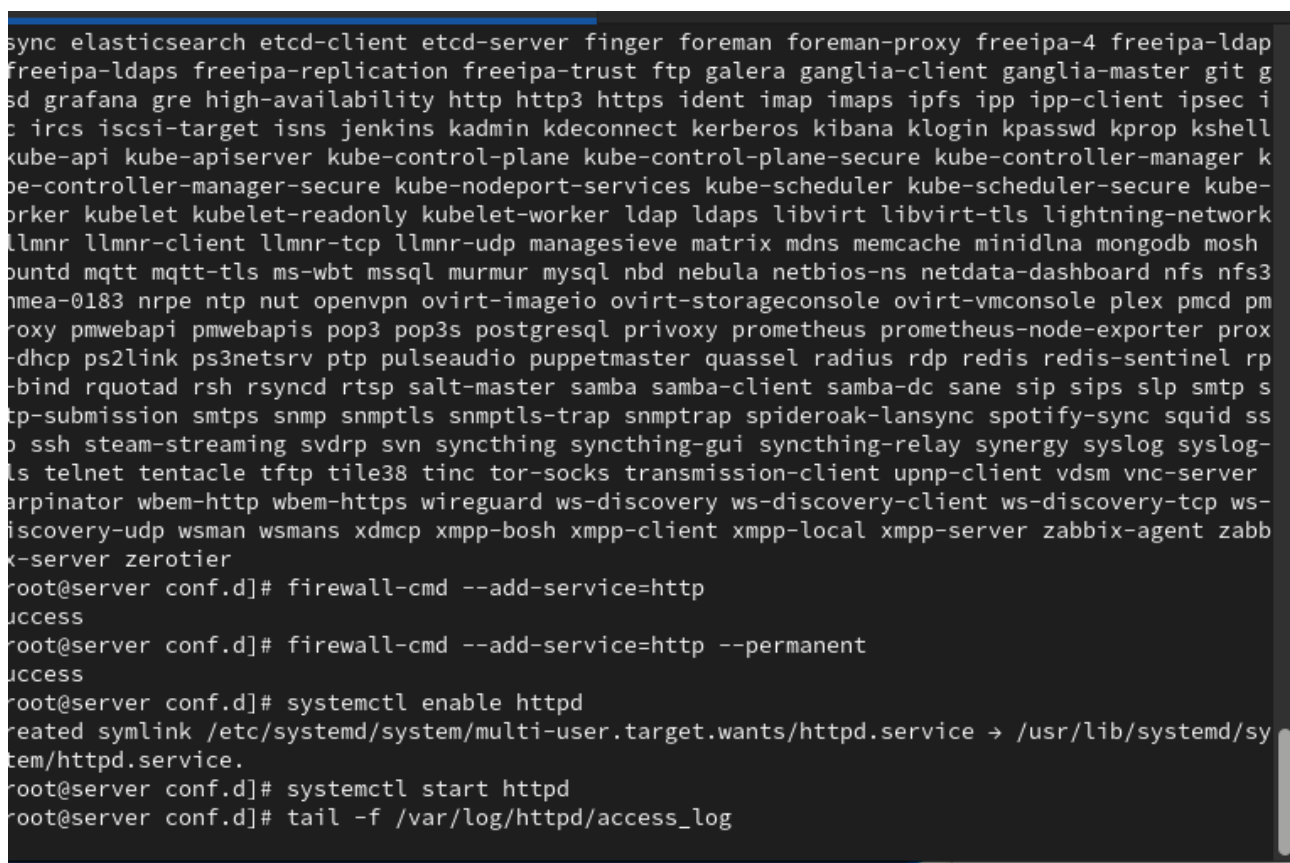
```
mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3
nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pm
proxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox
y-dhcp ps2link ps3netdrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rp
c-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp s
mtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ss
dp ssh steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-
tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsd vnc-server
warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-
discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabb
ix-server zerotier
[root@server conf.d]# firewall-cmd --add-service=http
success
[root@server conf.d]# firewall-cmd --add-service=http --permanent
success
[root@server conf.d]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/sy
stem/httpd.service.
[root@server conf.d]# systemctl start httpd
[root@server conf.d]#
```

Рис. 6: Этап 6



```
root@client:/  
[vagrant@client ~]$ sudo -i  
[root@client ~]# tail -f /var/log/httpd/error_log  
tail: cannot open '/var/log/httpd/error_log' for reading: No such file or directory  
tail: no files remaining  
[root@client ~]# cd ..  
[root@client /]# tail -f /var/log/httpd/error_log  
tail: cannot open '/var/log/httpd/error_log' for reading: No such file or directory  
tail: no files remaining  
[root@client /]#
```

Рис. 7: Этап 7



```
sync elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap  
freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git g  
sd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ipp-client ipsec i  
c ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell  
kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager k  
ube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-  
worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network  
lldmnr lldmnr-client lldmnr-tcp lldmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh  
nbd nebula netbios-ns netdata-dashboards nfs nfs3  
nfs4 nfs4-kerberos nfs4-ldap nfs4-ssl nfs4-ssh nfs4-ssl-kerberos nfs4-ssl-ldap nfs4-ssl-ssh nfs4-ssl-ssh-kerberos  
nfs4-ssl-ssh-kerberos-ldap nfs4-ssl-ssh-kerberos-ldap-ssl nfs4-ssl-ssh-kerberos-ldap-ssl-kerberos  
nfs4-ssl-ssh-kerberos-ldap-ssl-kerberos-ldap-ssl-kerberos-ldap-ssl-kerberos-ldap-ssl-kerberos-ldap-ssl-kerbos  
nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pm  
proxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox  
y-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rp  
sh-rbind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp s  
tp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ss  
h ssh steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-  
ls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsms vnc-server  
arpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-  
discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabb  
x-server zerotier  
root@server conf.d]# firewall-cmd --add-service=http  
success  
root@server conf.d]# firewall-cmd --add-service=http --permanent  
success  
root@server conf.d]# systemctl enable httpd  
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/sy  
stem/httpd.service.  
root@server conf.d]# systemctl start httpd  
root@server conf.d]# tail -f /var/log/httpd/access_log
```

Рис. 8: Этап 8

```

failed to start named.service: unit named.service not found.
[root@server ~]# cd /etc/httpd/conf.d
[root@server conf.d]# touch server.user.net.conf
[root@server conf.d]# touch www.user.net.conf
[root@server conf.d]# gedit user.net

(gedit:8297): dconf-WARNING **: 11:49:18.228: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:8297): dconf-WARNING **: 11:49:18.245: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:8297): dconf-WARNING **: 11:49:18.957: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:8297): dconf-WARNING **: 11:49:18.962: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:8297): dconf-WARNING **: 11:49:18.982: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

```

Рис. 9: Этап 9

Этап выполнения работы №10

```

(gedit:8297): dconf-WARNING **: 11:50:07.511: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server conf.d]# cd /var/www/html
[root@server html]# mkdir server.user.net
[root@server html]# cd /var/www/html/server.user.net
[root@server server.user.net]# touch index.html
[root@server server.user.net]# gedit

(gedit:8325): dconf-WARNING **: 11:51:15.509: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:8325): dconf-WARNING **: 11:51:15.512: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

```

Рис. 10: Этап 10

Этап выполнения работы №11

Этап выполнения работы №12

Этап выполнения работы №13

Этап выполнения работы №14



Рис. 11: Этап 11

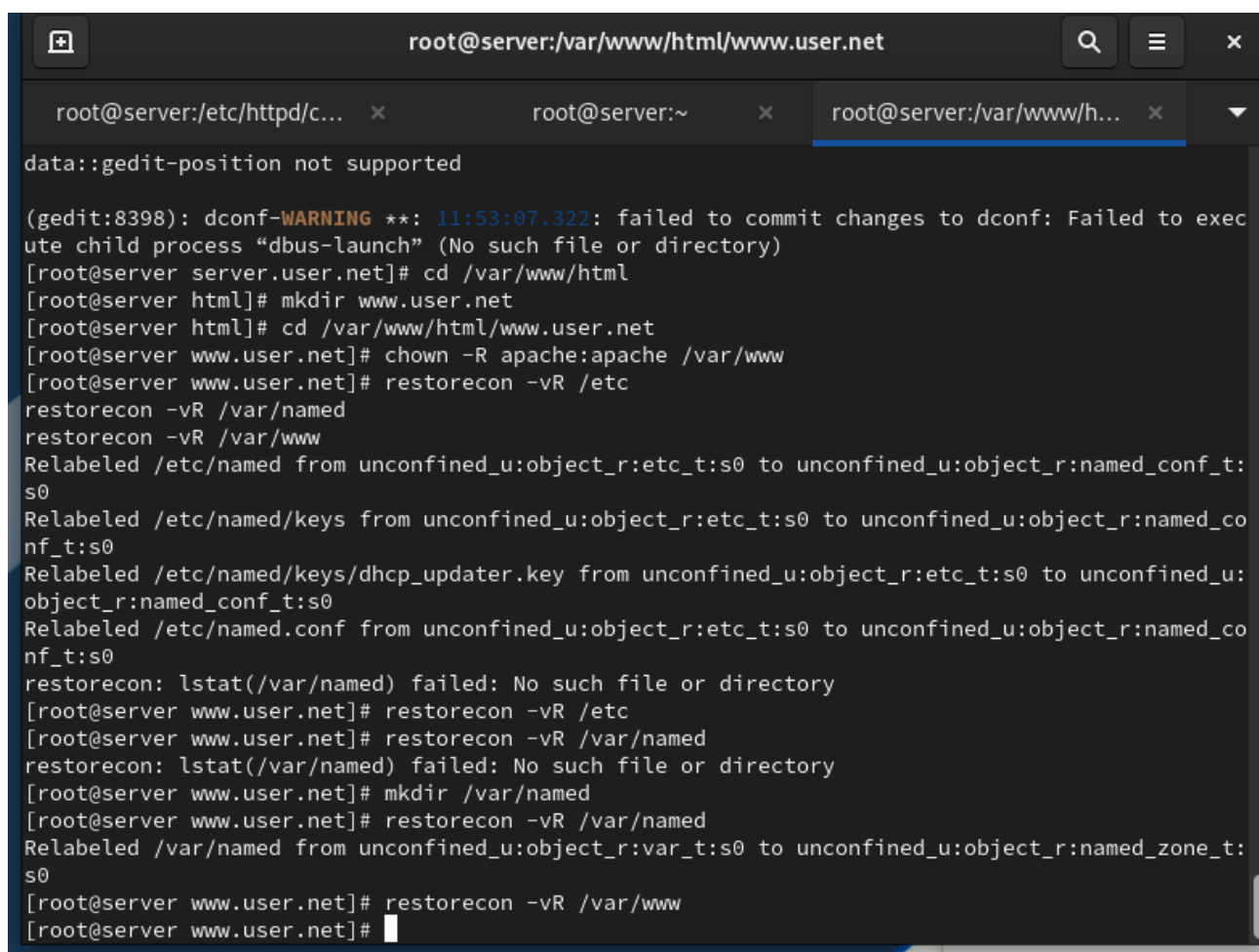


Рис. 12: Этап 12

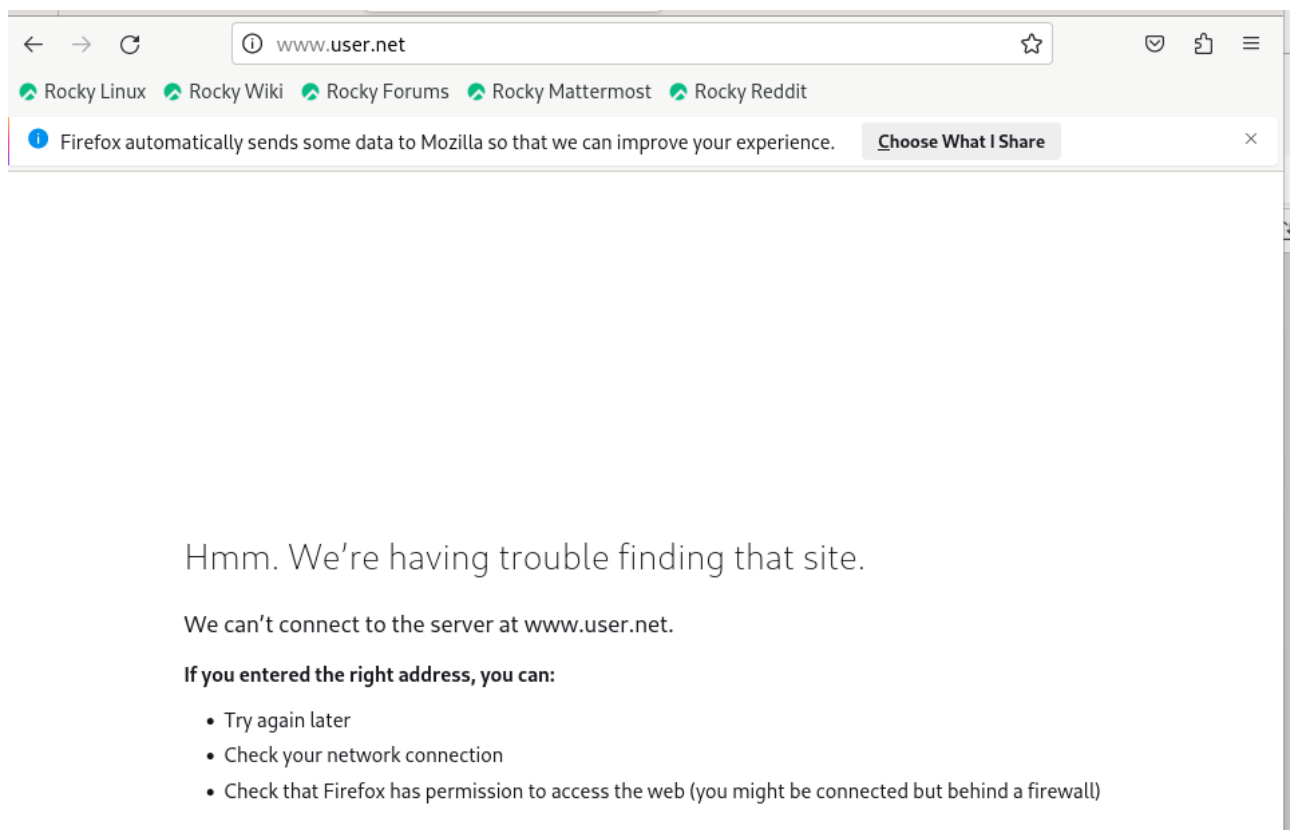


Рис. 13: Этап 13

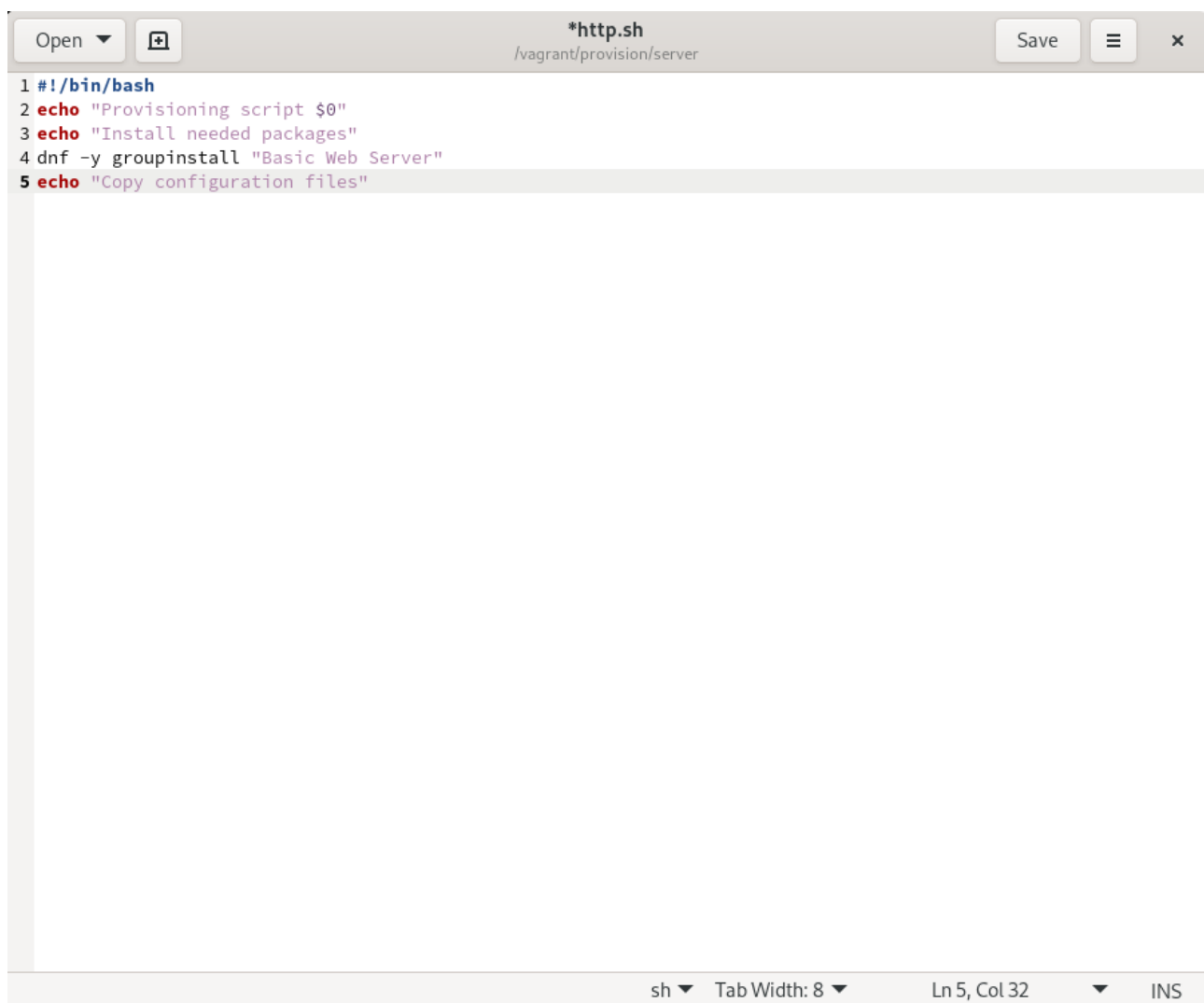


Рис. 14: Этап 14


```
root@server:/etc/httpd/c... x root@server:~ x root@server:/vagrant/pr... x
[root@server www.user.net]# restorecon -vR /var/named
Relabeled /var/named from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:named_zone_t:s0
[root@server www.user.net]# restorecon -vR /var/www
[root@server www.user.net]# systemctl restart httpd
[root@server www.user.net]# cd /vagrant/provision/server
-bash: cd: /vagrant/provision/server: No such file or directory
[root@server www.user.net]# cd .
[root@server www.user.net]# cd /
[root@server /]# cd /vagrant/provision/server
-bash: cd: /vagrant/provision/server: No such file or directory
[root@server /]# mkdir -p /vagrant/provision/server/http/etc/httpd/conf.d
[root@server /]# mkdir -p /vagrant/provision/server/http/var/www/html
[root@server /]# cp -R /etc/httpd/conf.d/*
/vagrant/provision/server/http/etc/httpd/conf.d/
cp: target '/etc/httpd/conf.d/www.user.net.conf' is not a directory
-bash: /vagrant/provision/server/http/etc/httpd/conf.d/: Is a directory
[root@server /]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d/
[root@server /]# cd /vagrant/provision/server/dns/
-bash: cd: /vagrant/provision/server/dns/: No such file or directory
[root@server /]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp: cannot stat '/var/named/*': No such file or directory
[root@server /]# cd /vagrant/provision/server
[root@server server]# touch http.sh
[root@server server]# chmod +x http.sh
[root@server server]# getdit http.sh
-bash: getdit: command not found...
[root@server server]# gedit http.sh
```

Рис. 15: Этап 15

Этап выполнения работы №15

Этап выполнения работы №16

Этап выполнения работы №17

Выводы

В ходе выполнения лабораторной работы были приобретены практические навыки

Спасибо за внимание!

Вопросы?

```
(gedit:8670): dconf-WARNING **: 12:04:22.291: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server server]# cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp: overwrite '/etc/httpd/conf.d/autoindex.conf'? yes
cp: overwrite '/etc/httpd/conf.d/fcgid.conf'? y
cp: overwrite '/etc/httpd/conf.d/manual.conf'? Y
cp: overwrite '/etc/httpd/conf.d/README'? Y
cp: overwrite '/etc/httpd/conf.d/rocky-snipolicy.conf'? all
cp: overwrite '/etc/httpd/conf.d/server.user.net.conf'?
cp: overwrite '/etc/httpd/conf.d/ssl.conf'?
cp: overwrite '/etc/httpd/conf.d/userdir.conf'?
cp: overwrite '/etc/httpd/conf.d/user.net'?
cp: overwrite '/etc/httpd/conf.d/welcome.conf'?
cp: overwrite '/etc/httpd/conf.d/www.user.net.conf'?
[root@server server]#
[root@server server]#
[root@server server]# cp -R /vagrant/provision/server/http/var/www/* /var/www
[root@server server]#
[root@server server]#
```

Рис. 16: Этап 16

```
root@server:/vagrant/provision/server
root@server:/etc/httpd/c... x root@server:~ x root@server:/vagrant/pr... x
cp: overwrite '/etc/httpd/conf.d/manual.conf'? Y
cp: overwrite '/etc/httpd/conf.d/README'? Y
cp: overwrite '/etc/httpd/conf.d/rocky-snipolicy.conf'? all
cp: overwrite '/etc/httpd/conf.d/server.user.net.conf'?
cp: overwrite '/etc/httpd/conf.d/ssl.conf'?
cp: overwrite '/etc/httpd/conf.d/userdir.conf'?
cp: overwrite '/etc/httpd/conf.d/user.net'?
cp: overwrite '/etc/httpd/conf.d/welcome.conf'?
cp: overwrite '/etc/httpd/conf.d/www.user.net.conf'?
[root@server server]#
[root@server server]#
[root@server server]# cp -R /vagrant/provision/server/http/var/www/* /var/www
[root@server server]#
[root@server server]# chown -R apache:apache /var/www
[root@server server]# restorecon -vR /etc
[root@server server]# restorecon -vR /var/www
[root@server server]# echo "Configure firewall"
Configure firewall
[root@server server]# firewall-cmd --add-service=http
Warning: ALREADY_ENABLED: 'http' already in 'public'
success
[root@server server]# firewall-cmd --add-service=http --permanent
Warning: ALREADY_ENABLED: http
success
[root@server server]# echo "Start http service"
Start http service
[root@server server]# systemctl enable httpd
[root@server server]# systemctl start httpd
[root@server server]#
```

Рис. 17: Этап 17